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Urban Land-Use Efficiency: A Stable Metric for SDG 11.3.1 and Its Economic and Structural Drivers

SDG 11.3.1: The Ratio of Land Consumption Rate to Population Growth Rate
193 UN Member Countries x 1985-2020 Panel Data Analysis

Ömer Furkan Çoban
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Prof. Dr. Jürgen Bitzer

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Introduction

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SDG 11.3.1: Ratio of Land Consumption Rate to Population Growth Rate

A planet in motion

Current World Population

8,304,854,184

• Live projection synchronized with Worldometers demographic parameters

TODAY	THIS YEAR
Births today 336,595	Births this year 71,062,514
Deaths today 158,605	Deaths this year 33,484,989
Population Growth today 177,990	Population Growth this year 37,577,525

Source: Live World Population Counter (Worldometer 2024).

- Every single year the world's built-up area grows by about the size of **the Netherlands**, as housing, roads and industry keep spreading outward (Seto et al. 2012).

But growth itself is not the problem

→ Cities **must** grow to house more people; that is normal.

→ What matters is **how much land each new resident uses**.

→ If land grows **at the same pace as** population, the city stays **compact**.

→ If land grows **much faster** than population, it **sprawls**.

→ So the problem is **sprawl, not growth**.

Why It Matters?: Sprawl is a development problem

Public Finance:
More land per person makes roads, water and transit cost far more per resident

Equity & Inclusion:
Sprawl pushes poorer residents to the periphery and fuels informal settlements

Land & Climate:
Sprawl permanently builds over farmland, locks cities into car dependence, and drives high-emission growth

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Measurement

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SDG Indicator 11.3.1: How Is It Calculated?

Computing 11.3.1, step by step (UN-Habitat 2025)

STEP 0: DEFINE THE ANALYSIS AREA: THE *URBAN* AREA

- Computed **inside cities**, not over whole countries
- Urban area delineated by the **Degree of Urbanisation (DEGURBA)** → **GHS-SMOD** grid
- Built-up & population below measured **only within urban cells**

STEP 1: MEASURE BUILT-UP AREA AND POPULATION AT TWO POINTS IN TIME

Where:

- V_{past}: Urban built-up (m²), start of period
- $V_{present}$: Urban built-up (m²), end of period
- t: Years in the built-up window (= 5)
- Pop_t: Urban population, start of period
- Pop_{t+n}: Urban population, end of period
- y: Years in the population window (= 5)

STEP 2: LAND CONSUMPTION RATE (LCR)

$$LCR = \frac{V_{present} - V_{past}}{V_{past}} \times \frac{1}{t}$$

How fast is the city's physical footprint expanding per year? → **Arithmetic (linear) rate**

STEP 3: POPULATION GROWTH RATE (PGR)

$$PGR = \frac{\ln(Pop_{t+n}/Pop_t)}{y}$$

How fast is the population growing per year? → **Logarithmic (exponential) rate**

STEP 4: THE RATIO

$$\boxed{LCRPGR = \frac{LCR}{PGR}}$$

How to read the result

LCRPGR < 1: Densification

Population grows faster than built-up area. More people per km² of city over time.

LCRPGR = 1: Balanced [SDG Target]

Land consumption and population grow at equal rates. Sustainable expansion.

LCRPGR > 1: Sprawl

Built-up area expands faster than population. Rising infrastructure costs per resident, farmland loss, transit failure.

A Flawed Yardstick: Two Structural Problems

Problem 1: Mathematical asymmetry

LCR uses an **arithmetic** (linear) rate; PGR uses a **logarithmic** (exponential) rate.

$$\underbrace{\frac{V_{present} - V_{past}}{V_{past}} \times \frac{1}{t}}_{\text{arithmetic LCR}} \div \underbrace{\frac{\ln(Pop_{t+n}/Pop_t)}{y}}_{\text{logarithmic PGR}}$$

A city with 5% annual growth in **both** built area and population does **not** produce LCRPGR = 1, the two "5%s" are on different mathematical scales. Taking a ratio of two differently-defined rates is dimensionally inconsistent: cross-country comparisons are distorted by the formula, not by real urbanisation.

Problem 2: Division instability

When $PGR \approx 0$ (shrinking or stable populations), LCRPGR diverges to $\pm\infty$.

Scenario	LCR	PGR	LCRPGR
Normal Growth	2%	1.5%	1.33
Shrinking Population	2%	-0.1%	**-20**
Stable Population	2%	0.001%	**2,000**

The same **2% land growth** reads as 1.33, -20, or 2,000 depending only on how close PGR is to zero. Because the denominator can vanish, the blow-up is **structural**: no winsorising or filtering repairs it, it only hides the symptom.

Worked example: 5%/yr in both, over 5 years

Same growth factor $g = 1.05^5 = 1.276$ for built area **and** population:

$$LCR = \frac{g-1}{5} = \frac{0.276}{5} = 5.53\%, \qquad PGR = \frac{\ln g}{5} = \frac{0.244}{5} = 4.88\%$$

$$\Rightarrow \mathbf{LCRPGR = 1.13} \text{ (not 1)}$$

Identical growth, yet the formula alone signals "sprawl" because $(g-1) > \ln g$.

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Built-up Area per Capita: The Metadata's Secondary Indicator

Recommended in the Official Metadata

The 11.3.1 metadata (UN-Habitat 2025) itself states LCRPGR is **hard to interpret** and proposes **two secondary indicators**, built from the *same inputs* (§4.f):

SECONDARY INDICATOR 1: BUILT-UP AREA PER CAPITA

$$\mathrm{BUP} = \frac{UrBU_t}{Pop_t} \quad (\mathrm{m}^2/\mathrm{person})$$

$\rightarrow$ Measures **land consumed per person**: the *state* at one point in time.

SECONDARY INDICATOR 2: TOTAL CHANGE IN BUILT-UP AREA

$$\text{Total change} = \frac{UrBU_{t+n} - UrBU_t}{UrBU_t} \quad (\%)$$

$\rightarrow$ Measures **total urban expansion** over the period (omits population).

We analyse the **growth rate of built-up area per capita**, the dynamic form of secondary indicator 1 (log form following (Lu and Weng 2026)):

$$\boxed{BpCR = \frac{\ln\left(\dfrac{UrBU_t/Pop_t}{UrBU_{t-z}/Pop_{t-z}}\right)}{z} = LCR_{\log} - PGR_{\log}}$$

$\rightarrow$ Measures the **annual trend in per-capita land use**: densifying (< 0) or sprawling (> 0).

Where:

- $UrBU_t$: urban built-up (m²) in year t
- Pop_t: urban population in year t
- z: period length in years (= 5)
- $t-z$ / $t+n$: earlier / later year
- $LCR_{\log} = \ln(UrBU_t/UrBU_{t-z})/z$: annual log growth of built-up
- $PGR_{\log} = \ln(Pop_t/Pop_{t-z})/z$: annual log growth of population

From a Level to a Trend

① **Built-up per capita(UN-Habitat 2025)**

$$\mathrm{BUP} = UrBU \,/\, Pop$$

The level: land per person, right now.

↓

② **Annual growth of BUP(Lu and Weng 2026)**

$$BpCR = \Delta \ln(\mathrm{BUP})/z = LCR_{\log} - PGR_{\log}$$

The change: is it rising or falling?

↓

③ **BpCR** (Our Dependent Variable)

▲ BpCR > 0	Land > People	**Sprawl**
◆ BpCR = 0	Land = People	**Balanced**
▼ BpCR < 0	People > Land	**Densification**

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Literature

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Literature Review: Drivers & Measurement

Drivers of Urban Expansion

A Meta-Analysis of Global Urban Land Expansion (Seto et al. 2011)

- **Data:** 326 remote-sensing studies, global, 1970–2000
- **Scope:** rates and drivers of global urban land expansion
- **Finding:** urban land grew **faster than population** almost everywhere; income drives it in most regions, population growth in India and Africa

Determinants of Urban Sprawl in European Cities (Oueslati et al. 2015)

- **Data:** panel of 282 European cities (1990, 2000, 2006)
- **Scope:** econometric determinants of sprawl (monocentric model)
- **Finding: rising income and population** expand the urban footprint; cities keep spreading even where population is flat

Measurement & SDG 11.3.1

Ratio of Land Consumption Rate to Population Growth Rate: Analysis of Different Formulations Applied to Mainland Portugal (Nicolau et al. 2019)

- **Data:** mainland Portugal, municipality level
- **Scope:** SDG 11.3.1: UN LCRPGR vs an LUE-derived formula
- **Finding:** the UN **LCRPGR formula is structurally unstable**; a log / LUE-based alternative is more robust

From Data Inconsistency to Reliable SDG 11.3.1 Monitoring: A Global Multi-Dataset Assessment of Land Use Efficiency (Lu and Weng 2026)

- **Data:** 6 built-up × 4 population datasets, 176 countries, 2000–2020
- **Scope:** reliability of LUE (11.3.1) across datasets
- **Finding:** LUE estimates **swing widely with dataset choice**; a **log-based** framework is needed for reliable monitoring

Our Contribution

- **Metric:** The **log-based BpCR** (Nicolau, Lu)
- **Method:** Make it **dynamic** with a two-way FE + **GMM(Generalized Method of Moments)** panel of **193 UN Member Countries (1985–2020)** measuring **persistence** (path-dependence)

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Data

Data Sources

GHSL

Global Human Settlement Layer

European Commission, Joint Research Centre / Copernicus

- **GHS-BUILT-S** (Built-up Surface), **GHS-POP** (Population), **GHS-SMOD** (Settlement Model)
- Built-up classified from **Landsat & Sentinel-2** by machine learning; GHS-POP disaggregates census onto that surface; **GHS-SMOD applies the Degree of Urbanisation (DEGURBA)** to flag *urban* cells
- **Available 1975–2030** at 5-year epochs (1975–2020 observed; 2025 & 2030 *modelled projections*)

European Union, Pesaresi et al. (2024); Schiavina et al. (2023)

GAUL 2024/2025

Global Administrative Unit Layers

FAO, Food and Agriculture Organization

- UN-validated administrative boundaries: politically **neutral**
- **Level 0** Country Level
- **Level 1** Province/Region Level
- **Global coverage** including all UN member states

Food and Agriculture Organization, FAO (2024)

UN

UN Statistics & Population Division

United Nations

- **Socio-economic controls** (built-up, population, urban share & density come from GHSL/SMOD)

Control	Source
GDP per capita Constant 2020 US$)	**UN SNAAMA**
Net migration (% of Population)	**UN DESA WPP 2024**

UN SNAAMA (2025); UN DESA WPP (2024); UN M49 / UN-CDP (2024)

Descriptive Statistics

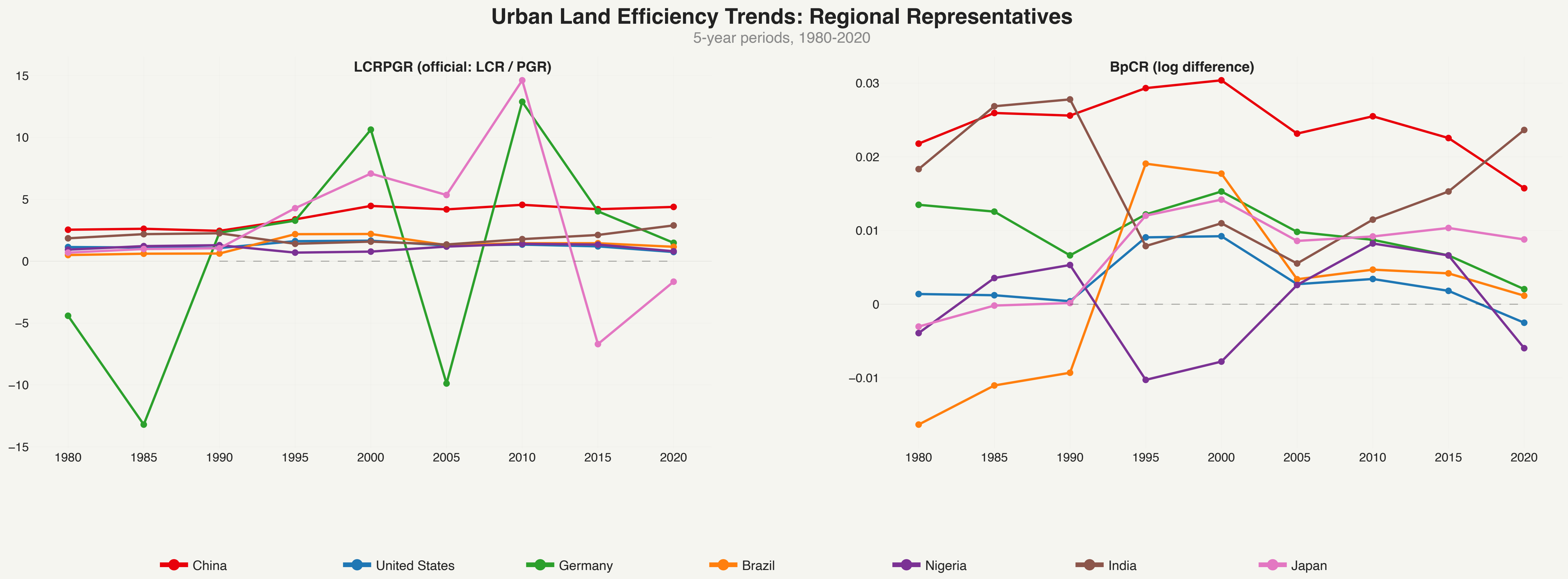
Descriptive Statistics

Variable	N	Mean	SD	Min	Median	Max
BpCR	1,500	0.004	0.017	−0.114	0.004	0.110
LCRPGR (log)	1,500	1.299	17.752	−567.915	1.025	281.447
ln(GDP per capita)	1,500	8.394	1.518	4.795	8.299	12.139
Urban Pop. Share[t−1]	1,500	0.662	0.171	0.103	0.690	1.000
ln(Urban Density)[t−1]	1,500	7.561	0.568	5.719	7.597	9.573
Net Migration (% Pop)	1,500	−0.059	1.121	−13.011	−0.051	13.200

Read the LCRPGR row

LCRPGR (log) already screams the problem: **SD ≈ 18** and a range from **-568 to +281**, vs **BpCR**'s SD ≈ **0.017** and range **[-0.11, +0.11]**. A ~1,000× wider spread: driven by LCRPGR's division instability when PGR ≈ 0.

Country Trends: Regional Representatives



i **Key patterns**

- **China**: strongest, most persistent per-capita sprawl (BpCR ≈ +0.02 to +0.03), peaking around 2000 then easing to 0.016; LCRPGR drifts up to ~4.4.
- **Germany**: trends toward balance (BpCR 0.015 → 0.002), yet its LCRPGR swings wildly (−13.2 to +12.9).
- **Japan**: BpCR negative in the 1980s, then a low, stable ~+0.01, while its LCRPGR lurches from +14.6 to -6.7.
- **Nigeria & Brazil**: oscillate around zero in BpCR (no persistent sprawl), with low LCRPGR (~1).
- **Takeaway**: BpCR gives a stable, interpretable per-capita signal; LCRPGR does not.

Empirical Analysis

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Research Question & Hypotheses

Research question

What economic and structural factors drive within-country changes in per-capita urban land use over time, and how persistent are these trajectories?

> **Measurement sub-question**
>
> Does the official SDG 11.3.1 indicator (**LCRPGR**) reliably capture this, or does the **choice of metric** change the conclusions?

Hypotheses

	Hypothesis	Expected Sign
H1	BpCR measures efficiency better than LCRPGR	Higher Fit
H2	Higher urban density → less land/capita	–
H3	Higher income → less land/capita	–
H4	Net in-migration densifies cities	–
H5	Per-capita land use is path-dependent	+

Empirical Strategy

One specification, two dependent variables

$$\text{Metric}_{i,t} = \rho\,\text{Metric}_{i,t-1} + \beta_1 \ln(UrbDens)_{i,t} + \beta_2\, UrbShare_{i,t} + \beta_3 \ln(GDPpc)_{i,t} + \beta_4\, NetMigr_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}$$

Two-stage plan

① Compare the metric **statically** (drop $\rho\,\text{Metric}_{i,t-1}$) to isolate **LCRPGR vs BpCR**

② **Add the lagged DV** for dynamics and address its Nickell bias with **GMM** *(Final Model)*

The same model is estimated twice; only Y changes:

	Model 1	Model 2
DV	**LCRPGR** (UN)	**BpCR** (Ours)
FE	Country + Period	Country + Period
SE	Clustered (country)	Clustered (country)
N	1499	1499

- **Dynamic two-way FE panel**: country + period fixed effects ($\mu_i,\ \delta_t$)
- **Same RHS, two DVs** (LCRPGR vs BpCR): any divergence isolates the *metric*, not the data
- **Lagged DV** ($\rho\, Y_{t-1}$): captures inertia; the FE ρ is Nickell-biased downward, GMM-corrected (see Final Model)
- **Controls** (contemporaneous, at t): urban density, urban share, GDP pc, net migration
- **193 countries** $\times$ **8 five-year periods** (1985–2020); SE clustered by country

Why Each Variable Enters

Variables	Rationale
Lagged DV (ρ)	Captures inertia: once a country sprawls, it tends to keep sprawling
ln(Urban Density)	Urban form: denser cities consume less new land per resident
Urban Pop. Share	How urbanised the country already is; less rural land left to convert
ln(GDP Per Capita)	Tests whether richer countries use more or less built-up land per person
Net Migration	Demographic pressure: do newcomers densify cities or push outward expansion?
Country & Period FE	Absorb fixed geography and institutions, plus shocks shared in each period

Model 1: UN Approach: LCRPGR Regression

$$LCRPGR_{i,t} = \beta_1 \ln(UrbDens)_{i,t} + \beta_2\, NetMigr_{i,t} + \beta_3 \ln(GDPpc)_{i,t} + \beta_4\, UrbShare_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}$$

	Dependent variable: LCRPGR (log) (193 countries × 8 periods, 1985–2020)			
	Density	+ Net Migr.	+ GDP pc	+ Urban %
ln(Urban Density)[t−1]	-3.107	-3.143	-1.979	-0.772
	(2.907)	(2.930)	(2.647)	(2.212)
Net Migration (% Pop)		-0.142	-0.215	-0.261
		(0.184)	(0.200)	(0.234)
ln(GDP per capita)			2.853*	2.867*
			(1.720)	(1.729)
Urban Pop. Share[t−1]				-11.808
				(13.099)
Observations	1499	1499	1499	1499
Within R^2	0.000	0.000	0.002	0.003
Country FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes

* p < 0.1, ** p < 0.05, *** p < 0.01

SE clustered at country level. DV: LCRPGR (log) = Land Consumption Rate ÷ Population Growth Rate (both log rates). Static two-way FE (no lagged DV); GHSL/SMOD + UN (GDP, net migration).

Model 2: Our Approach: BpCR Regression

$$BpCR_{i,t} = \beta_1 \ln(UrbDens)_{i,t} + \beta_2\, NetMigr_{i,t} + \beta_3 \ln(GDPpc)_{i,t} + \beta_4\, UrbShare_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}$$

	Dependent variable: BpCR (193 countries × 8 periods, 1985–2020)			
	Density	+ Net Migr.	+ GDP pc	+ Urban %
ln(Urban Density)[t−1]	0.0148**	0.0132**	0.0118**	0.0103*
	(0.0062)	(0.0060)	(0.0057)	(0.0059)
Net Migration (% Pop)		-0.0066***	-0.0065***	-0.0064***
		(0.0012)	(0.0013)	(0.0013)
ln(GDP per capita)			-0.0032	-0.0032
			(0.0023)	(0.0023)
Urban Pop. Share[t−1]				0.0147
				(0.0215)
Observations	1499	1499	1499	1499
Within R^2	0.015	0.207	0.211	0.212
Country FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

SE clustered at country level. DV: BpCR = Δln(built-up per capita)/z ≡ LCR − PGR. Static two-way FE (no lagged DV); GHSL/SMOD + UN (GDP, net migration).

Final Model: BpCR under Dynamic-Panel GMM

$$BpCR_{i,t} = \rho\, BpCR_{i,t-1} + \beta_1 \ln(UrbDens)_{i,t} + \beta_2\, NetMigr_{i,t} + \beta_3 \ln(GDPpc)_{i,t} + \beta_4\, UrbShare_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t}$$

The dynamic FE model is built up control-by-control (first five columns); the last two columns target the lagged-DV **Nickell bias** with **Arellano-Bond** and **Blundell-Bond** GMM (instrumenting the lagged BpCR with its deeper lags).

Dependent variable: BpCR

Δln(built-up area ÷ population) / z ≡ log-LCR – log-PGR

(193 countries × 8 periods, 1985–2020)

	Fixed Effects					GMM (Bias-Corrected)	
	Base	+ Density	+ Net Migr.	+ GDP	+ Urban %	Arellano-Bond	Blundell-Bond
BpCR[t−1]	+0.2535*** (0.0578)	+0.2670*** (0.0555)	+0.2584*** (0.0497)	+0.2565*** (0.0501)	+0.2558*** (0.0505)	+0.2129** (0.1032)	+0.4554*** (0.1059)
ln(Urban Density)[t−1]		+0.0182*** (0.0050)	+0.0165*** (0.0053)	+0.0154*** (0.0049)	+0.0142*** (0.0051)	+0.0371*** (0.0141)	-0.0010 (0.0015)
Net Migration (% Pop)			-0.0065*** (0.0011)	-0.0064*** (0.0011)	-0.0064*** (0.0012)	-0.0062*** (0.0022)	-0.0052*** (0.0011)
ln(GDP per capita)				-0.0026 (0.0022)	-0.0026 (0.0022)	-0.0061 (0.0044)	+0.0007 (0.0006)
Urban Pop. Share[t−1]					+0.0112 (0.0202)	+0.0350 (0.0778)	+0.0087 (0.0063)
Observations	1,499	1,499	1,499	1,499	1,499	1,351	1,544
Within R²	0.076	0.099	0.285	0.287	0.288	–	–
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Instruments	–	–	–	–	–	15	21
Hansen (p)	–	–	–	–	–	0.05	0.01
AR(1) (p)	–	–	–	–	–	0.01	0.02
AR(2) (p)	–	–	–	–	–	0.76	0.62

*** p<0.10, ** p<0.05, *** p<0.01.**
Standard errors clustered by country.
Notes:
1) Cols 1–5: dynamic two-way FE (country + period FE), controls added stepwise.
2) AB = Arellano-Bond difference GMM; **BB** = Blundell-Bond system GMM, instrumenting the lagged BpCR with lags ≥ 2 (two-step, Windmeijer-robust SE).
3) Hansen J = over-identification test (p > 0.10 → valid instruments); **AR(1)/AR(2)** = first/second-order serial correlation (AR(1) significant by construction; AR(2) p > 0.10 → valid).

BpCR[t−1] INERTIA

ρ = +0.21** Here ρ is imprecise (SE 0.10) and its moments are marginal, so we do not read it: it sits *just* below the Bond bracket, within sampling error. Where the moments are clean: **ρ ≈ 0.46 (SE 0.07)** — Nickell-implied 0.45, AB 0.46, BB 0.45 all agree.

H5 ✓ Supported: ~half of last period's BpCR persists

ln(Urban Density)[t−1] SIGN FLIPS

β_1 = +0.037*** BpCR contains **−ln $P_{(t)}$ + ln $P_{(t-1)}$**, so density inherits the sign of its own date: **−0.055***** at *t*, **positive** at *t−1*. Clean *t−2* timing: **+0.011****.

H2 ✗ Rejected: the compact-city sign was arithmetic

Net Migration DENSIFICATION

β_2 = −0.0062*** **Our strongest result.** Migration moves with urban population **+0.88 pp/yr** but built-up only **+0.24** → the gap is densification. Holds in **all 3** development groups.

H4 ✓ Supported as an *association*: passes a strict-exogeneity test, but migrants may select into fast-building economies

ln(GDP per capita) NOT SIGNIFICANT

β_3 = −0.006 (n.s.) Sign is negative in **every** spec and group, but never significant once migration is measured over the right window.

H3 ✗ Not supported: consistent sign, insufficient precision

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Conclusion

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Conclusion

- **Sprawl is real and measurable.** In most world regions urban land grows faster than population: a policy-relevant inefficiency.
  - BpCR > 0 (sprawl) in **63%** of country-period observations; **67% of countries (129/193)** sprawl on average over time (Source: own calculations (GHSL/SMOD + UN data, 1985–2020))
  - Median BpCR = **+0.0041** (built-up per capita rising); LCRPGR > 1 in **56%** of cases (Source: own calculations (GHSL/SMOD + UN data, 1985–2020))
- **The official LCRPGR is unreliable.** A raw ratio that explodes when population growth is near zero (division instability) and explains almost nothing within countries (within $R^2 \approx 0.003$).
- **Our log-based BpCR is the better metric.** Symmetric and numerically stable (following Nicolau and Lu), it fits about **20× better** (within $R^2 \approx 0.065$ static, ≈ 0.13 dynamic).
- **Robust drivers (Bias-Corrected GMM).** Denser cities densify (negative, highly significant); sprawl is **path-dependent** ($\rho \approx 0.6$); net migration densifies; and **higher income densifies too** (negative, significant).
- **We complement the literature, not contradict it.** Between countries, richer ones are more sprawled (the usual finding); within a single country the income sign turns negative, but it is not precisely estimated. What our country fixed effects *do* recover robustly is the **migration** channel: cities absorb in-migrants about **4× faster than they build for them**, so in-migration means densification, in developed, developing and least-developed economies alike.
- **Methodological contribution.** A dynamic two-way FE + **GMM** panel of **193 UN member countries (1985–2020)** that targets the Nickell bias and brackets persistence.

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Thank you for your attention!

Questions?

Appendix: Backup Slides

The Five Indicators at a Glance

Indicator	Formula (annualised)	Type	What it measures	Caveat
LCR · Land Consumption Rate	$\frac{UrBU_t - UrBU_{t-z}}{UrBU_{t-z}} \cdot \frac{1}{z}$ *(arithmetic)*	rate	how fast built-up area expands	ignores population
PGR · Population Growth Rate	$\ln(Pop_t / Pop_{t-z})/z$ *(log)*	rate	how fast urban population grows	building block
LCRPGR · *official 11.3.1*	LCR/PGR	ratio	land growth **relative to** population growth	unstable: $\to \pm\infty$ as $PGR \to 0$; arithmetic $\div$ log
BUP · Built-up per capita *(secondary 1)*	$UrBU_t / Pop_t$	level	land consumed per person (the *state*)	a stock: mechanical as a regression DV
BpCR · *our dependent variable*	$\frac{\ln(BUP_t/BUP_{t-z})}{z} = LCR_{\log} - PGR_{\log}$	rate	the *trend* in per-capita land use (efficiency)	not named in metadata; growth of secondary 1

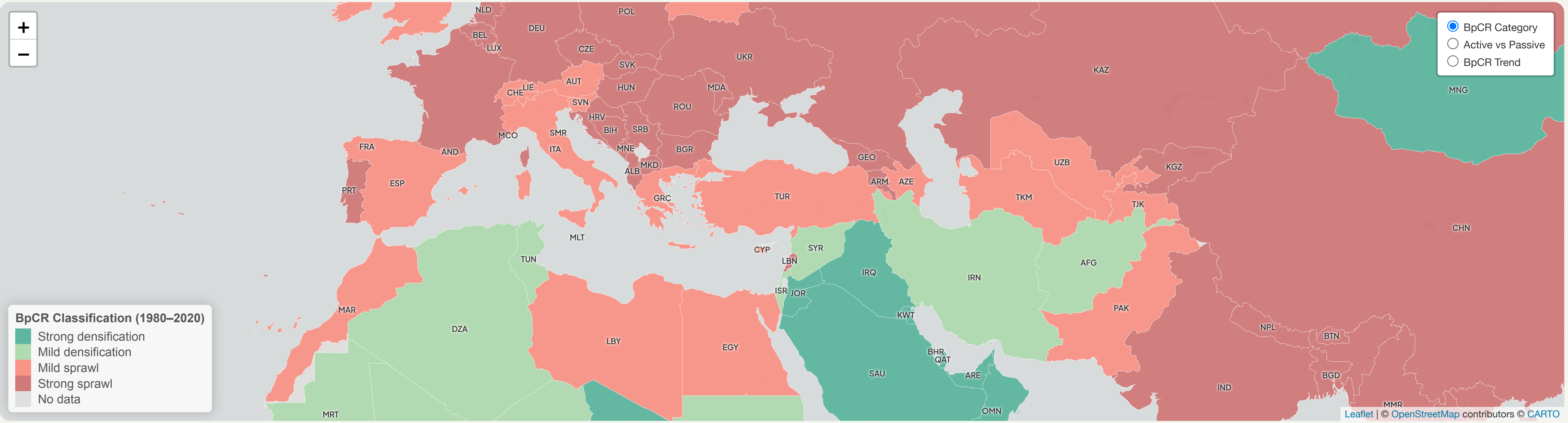
Metadata secondary indicator 2, Total change in built-up = $(UrBU_{t+n} - UrBU_t)/UrBU_t$, measures total expansion but, like LCR, omits population. $UrBU$ = urban built-up area, Pop = urban population, z = period length (5 yr).

Carl von Ossietzky Universität Oldenburg

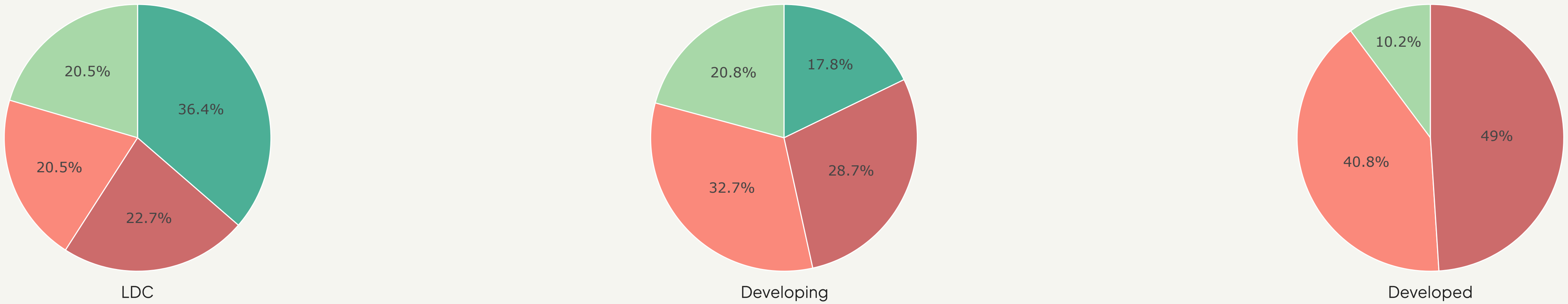
Global BpCR Distribution

Urban Land Consumption Efficiency: Global Distribution

Category: 1980–2020 | Trend: 2000–2020



Distribution by Development Group (%)



Strong densification: BpCR < -0.007 **Mild densification:** -0.007 ≤ BpCR < 0 **Mild sprawl:** 0 ≤ BpCR < 0.009 **Strong sprawl:** BpCR ≥ 0.009

Country mean BpCR, 1980–2020. Thresholds: 0 (theory) and the median of each side.

11 SUSTAINABLE CITIES AND COMMUNITIES